ASHUTOSH SINGH

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CAREER SUMMARY:

Full Stack & MES Developer with 3.5+ years of experience engineering scalable manufacturing and web solutions at Tata Technologies. Skilled in React, Node.js, Python/Django, .NET, and Apriso/FTPC systems. Proven track record in automating shop-floor operations, optimizing REST APIs, and delivering real-time Production dashboards.

WORK EXPERIENCE:

Solution Developer | Tata Technologies
Apr 2024 – Present

- **DELMIA Apriso Development:** Developed custom Apriso applications, business logic and reporting solutions tailored to Production and Quality module requirements; implemented key process changes to optimize shop-floor workflows and tracking.
- **BIW Shop MES Application:** Architected an end-to-end MES web application using **React, Node.js, Express, and MSSQL;** digitized shop-floor operations across order start, booking, punching, and defect logging.
- **CORAL MES (Rockwell FTPC):** Engineered 10+ FTPC and Java-based manufacturing applications for the CORAL line; designed role-based workflows and process validations.
- **Real-Time Dashboards & Alerts:** Developed live monitoring portals with integrated **KEPServer APIs** and automated voice alerts, drastically improving incident response time for line stoppages.
- **Backend & Automation:** Automated BOM validation workflows using SQL Server stored procedures and built ASP.NET Core/C# API integrations.
- **Performance Optimization:** Optimized complex SQL queries, indexes and backend API endpoints, reducing average data fetch times and dashboard latency by ~30%.

Full Stack Developer | TechPath Research & Development
Jul 2023 – Feb 2024

- **Backend Architecture:** Built scalable backend architectures using **Django & DRF**, including e-commerce engines (cart, variations, Razorpay) and a role-based Project Management System.
- **Frontend Development:** Developed frontend interfaces in **React.js** with real-time state management and optimized REST API communication.
- **API Security & Integration:** Designed and integrated secure JWT authentication mechanisms and third-party webhooks to maintain robust data exchange across client platforms.
- **Code Quality & Maintenance:** Refactored legacy frontend components into modular, reusable React components, significantly improving code maintainability and development velocity.

Python Intern | TechPath Research & Development
Dec 2022 – Jun 2023

- **Web Application Development:** Developed Flask web apps with SQLAlchemy ORM, managing authentication, session security, and database query optimization.
- **Database & Query Tuning:** Wrote optimized raw SQL queries and custom ORM filters to minimize database load and prevent redundant data queries during peak traffic.

SKILLS:

- **Languages & Framework:** C#, Python, JavaScript, Java, React.js, Next.js, Node.js, Express.js, Django, Flask, ASP.NET Core, React Native, Angular
- **Databases & Tools:** MSSQL Server, SQL, Git, Azure, KEPServer, SQLAlchemy, REST APIs
- **Domain Expertise:** Manufacturing Execution System (MES), Apriso, Rockwell FTPC

EDUCATION:

Master's of Computer Application (MCA)
Manipal University

09/2023 - 11/2025 **Jaipur, India**

Bachelor's of Computer Application (BCA)
Mahatma Gandhi Kashi Vidyapeeth

11/2020 - 09/2023 **Varanasi, India**